

Short Interval Shutter

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

This document describes the function that enables continuous exposure at intervals of 2 μ s or more.

This function can be used to expose two consecutive frames to exposure lights respectively in a short interval, for example, for Particle Image Velocimetry (PIV) which is used for detection of particles or measurement of particle velocity.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

3. Features

◆ Frame identifier

Frame information line (In case of SLVS output)

Embedded data line (In case of SLVS – EC output or CSI-2 serial data output)

◆ Pulse output

Output exposure period control signal or exposure prohibited period control signal to TOUT0, TOUT1 and TOUT2

◆ Support mode

Normal mode (XVS / XHS sync)

* Trigger modes are not supported.

◆ Supported readout drive mode

All-pixel scan mode

Vertical / Horizontal 1 / 2 Subsampling mode

2 × 2 FD binning mode (The product with a color filter is excluded)

ROI mode

Overlap ROI mode

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

◆ Support each frame gain adjust function

Contents

1.	Product type	1
2.	Description	1
3.	Features	2
	Contents	3
4.	Register Map	4
5.	Short Interval Shutter mode	21
5.1.	Overview	21
6.	Setting of Short Interval Shutter	22
6.1.	Exposure Period	23
6.2.	Exposure Interval and Exposure period	24
6.2.1.	Exposure prohibited control signal	24
6.2.2.	Exposure period control signal	25
6.2.3.	Relationship between exposure time and control signals	26
6.2.4.	Pulse Output Function	27
7.	ROI Function	29
7.1.	In case of ROI	29
7.2.	In case of Overlap ROI	31
8.	Reflection Timing	33
8.1.	Timing of the 1 st exposure period change	33
9.	Each frame Gain mode	34
9.1.	Gain reflection timing	35
10.	Frame identification method	36
10.1.	Frame information line	36
10.2.	Output format (Frame information line)	36
10.3.	Embedded data line	37
10.4.	Output format (Embedded data line)	37
11.	Mode Transitions	39
11.1.	Mode Transitions	39
12.	Limitations	39
	Revision History	40

4. Register Map

The register map for the short interval shutter is shown below.
Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 02h in Write mode. (I²C:30**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
E0h	30E0h	0	VNDMY [7:0]	Setting of the 1 st exposure period VNDMY + GMRWT + GMTWT (unit: H) is the 1 st available exposure period, and the exposure time can be adjusted within this period. Set the value of multiple of 4. (Set the value of multiple of 8 when using 1/2 subsampling mode or 2x2 FD binning mode with a monochrome sensor.)	04h	04h	S
		1					
		2					
		3					
		4					
		5					
		6					
		7					

Registers corresponding to Chip ID = 03h in Write mode. (I²C:31**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3100h	0	ROI_MODE [0]	Selection of ROI mode 0: ROI Mode 1: Overlap ROI Mode	0	00h	S
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
04h	3104h	0	FID0_ROIH1ON [0]	The horizontal setting of FID0 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID0_ROIV1ON [0]	The vertical setting of FID0 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID0_ROIH2ON [0]	The horizontal setting of FID0 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID0_ROIV2ON [0]	The vertical setting of FID0 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID0_ROIH3ON [0]	The horizontal setting of FID0 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID0_ROIV3ON [0]	The vertical setting of FID0 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID0_ROIH4ON [0]	The horizontal setting of FID0 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID0_ROIV4ON [0]	The vertical setting of FID0 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
06h	3106h	0	FID1_ROIH1ON [0]	The horizontal setting of FID1 ROI area (1, y) (y = 1 to 4) 0: Disable 1: Enable	0	00h	V
		1	FID1_ROIV1ON [0]	The vertical setting of FID1 ROI area (x, 1) (x = 1 to 4) 0: Disable 1: Enable	0		I
		2	FID1_ROIH2ON [0]	The horizontal setting of FID1 ROI area (2, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		3	FID1_ROIV2ON [0]	The vertical setting of FID1 ROI area (x, 2) (x = 1 to 4) 0: Disable 1: Enable	0		I
		4	FID1_ROIH3ON [0]	The horizontal setting of FID1 ROI area (3, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		5	FID1_ROIV3ON [0]	The vertical setting of FID1 ROI area (x, 3) (x = 1 to 4) 0: Disable 1: Enable	0		I
		6	FID1_ROIH4ON [0]	The horizontal setting of FID1 ROI area (4, y) (y = 1 to 4) 0: Disable 1: Enable	0		V
		7	FID1_ROIV4ON [0]	The vertical setting of FID1 ROI area (x, 4) (x = 1 to 4) 0: Disable 1: Enable	0		I
20h	3120h	0	FID0_ROIPH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
21h	3121h	7		Designation of horizontal cropping position for FID0 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID0_ROIPV1 [12:0]	MSB	0	00h	—
		1					
		2					
		3					
		4					
		5		Fixed to 0			
		6		Fixed to 0			
22h	3122h	7		Fixed to 0			
		0	FID0_ROIPH1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
23h	3123h	7		Designation of vertical cropping position for FID0 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID0_ROIPV1 [12:0]	MSB	0	00h	—
		1					
		2					
		3					
		4					
		5		Fixed to 0			
		6		Fixed to 0			
23h	3123h	7		Fixed to 0			

FID0_ROI WV1 [12:0]	LSB
	Designation of vertical cropping size for FID0 area (x, 1) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}
	MSB
	Fixed to 0
	Fixed to 0
	Fixed to 0
	LSB

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
2Ah	312Ah	0	FID0_ROIPV2 [12:0]	LSB	0000h	00h	I	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
2Bh	312Bh	0		MSB	0	00h	—	
		1						
		2						
		3						
		4						
		5		Fixed to 0				0
		6		Fixed to 0				0
		7		Fixed to 0				0
2Ch	312Ch	0	FID0_ROIWH2 [12:0]	LSB	0000h	00h	V	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
2Dh	312Dh	0		MSB	0	00h	—	
		1						
		2						
		3						
		4						
		5		Fixed to 0				0
		6		Fixed to 0				0
		7		Fixed to 0				0
2Eh	312Eh	0	FID0_ROIWH2 [12:0]	LSB	0000h	00h	I	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
2Fh	312Fh	0		MSB	0	00h	—	
		1						
		2						
		3						
		4						
		5		Fixed to 0				0
		6		Fixed to 0				0
		7		Fixed to 0				0

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
30h	3130h	0	FID0_ROIPH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
31h	3131h	7		Designation of horizontal cropping position for FID0 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²			
		0					
		1					
		2					
		3					
		4					
		5					
32h	3132h	6	FID0_ROIPV3 [12:0]	MSB	0000h	00h	I
		7					
		0					
		1					
		2					
		3					
		4					
33h	3133h	5		Fixed to 0	0		—
		6					
		7					
		0					
		1					
		2					
		3					
34h	3134h	4	FID0_ROIWH3 [12:0]	Fixed to 0	0000h	00h	V
		5					
		6					
		7					
		0					
		1					
		2					
35h	3135h	3		Designation of horizontal cropping size for FID0 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	—
		4					
		5					
		6					
		7					
		0					
		1					
		2		MSB	0		—
		3					
		4					
		5					
		6					
		7					
		0					

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
36h	3136h	0	FID0_ROI WV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
37h	3137h	7	FID0_ROI WV3 [12:0]	Designation of vertical cropping size for FID0 on area (x, 3) (x = 1 to 8) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
		3					
		4					
		5					
38h	3138h	6	FID0_ROI PH4 [12:0]	Designation of horizontal cropping position for FID0 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	V
		7					
		0					
		1					
39h	3139h	2	FID0_ROI PH4 [12:0]	Designation of horizontal cropping position for FID0 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	V
		3					
		4					
		5					
		6					
		7					
		0					
3Ah	313Ah	1	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		2					
		3					
		4					
		5					
		6					
		7					
3Bh	313Bh	0	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		0					
		1					
		2					
3Bh	313Bh	3	FID0_ROI PV4 [12:0]	Designation of vertical cropping position for FID0 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16* ¹ *Set the value of multiple of 8* ²	0000h	00h	I
		4					
		5					
		6					
3Bh	313Bh	7	FID0_RO				

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
3Ch	313Ch	0	FID0_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
3Dh	313Dh	7		Designation of horizontal cropping size for FID0 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
3Eh	313Eh	0	FID0_ROIWH4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
3Fh	313Fh	7		Designation of vertical cropping size for FID0 on area (x, 4) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
60h	3160h	0	FID1_ROIPH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
61h	3161h	7		Designation of horizontal cropping position for FID1 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
62h	3162h	0	FID1_ROIPV1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
63h	3163h	7		Designation of vertical cropping position for FID1 on area (x, 1) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
64h	3164h	7		Fixed to 0			
		0	FID1_ROIWH1 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
65h	3165h	7		Designation of horizontal cropping size for FID1 on area (1, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
66h	3166h	7		Fixed to 0			
		0	FID1_ROIWW1 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
67h	3167h	7		Designation of vertical cropping size for FID1 on area (x, 1) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
67h	3167h	7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
68h	3168h	0	FID1_ROIPH2 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
69h	3169h	7		Designation of horizontal cropping position for FID1 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROIPV2 [12:0]		0000h	00h	I
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Ah	316Ah	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
6Bh	316Bh	7		Designation of vertical cropping position for FID1 on area (x, 2) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Ch	316Ch	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
6Dh	316Dh	7		Designation of horizontal cropping size for FID1 on area (2, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			
		0	FID1_ROIWH2 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
6Dh	316Dh	7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
6Eh	316Eh	0	FID1_ROI WV2 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
6Fh	316Fh	7		Designation of vertical cropping size for FID1 on area (x, 2) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROI PH3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
70h	3170h	7		Fixed to 0			
		0	FID1_ROI PV3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
71h	3171h	7		Designation of horizontal cropping position for FID1 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROI WV3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
72h	3172h	7		Fixed to 0			
		0	FID1_ROI PH3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
73h	3173h	7		Designation of vertical cropping position for FID1 on area (x, 3) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0	FID1_ROI WV3 [12:0]		0000h	00h	V
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
73h	3173h	7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
74h	3174h	0	FID1_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
75h	3175h	7		Designation of horizontal cropping size for FID1 on area (3, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
76h	3176h	7		Fixed to 0			
		0	FID1_ROIWH3 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
77h	3177h	7		Designation of vertical cropping size for FID1 on area (x, 3) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
78h	3178h	7		Fixed to 0			
		0	FID1_ROIWH3 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
79h	3179h	7		Designation of horizontal cropping position for FID1 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
79h	3179h	7		Fixed to 0			

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
7Ah	317Ah	0	FID1_ROIPV4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
7Bh	317Bh	7		Designation of vertical cropping position for FID1 on area (x, 4) (x = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
7Ch	317Ch	7		Fixed to 0			
		0	FID1_ROIWH4 [12:0]	LSB	0000h	00h	V
		1					
		2					
		3					
		4					
		5					
		6					
7Dh	317Dh	7		Designation of horizontal cropping size for FID1 on area (4, y) (y = 1 to 4) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
7Eh	317Eh	7		Fixed to 0			
		0	FID1_ROIWH4 [12:0]	LSB	0000h	00h	I
		1					
		2					
		3					
		4					
		5					
		6					
7Fh	317Fh	7		Designation of vertical cropping size for FID1 on area (x, 4) (x = 1 to 8) *Set the value of multiple of 16 ^{*1} *Set the value of multiple of 8 ^{*2}			
		0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

^{*1} IMX530 series and IMX535 series.

^{*2} IMX540 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Registers corresponding to Chip ID = 05h in Write mode. (I²C:33**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
A4h	33A4h	0	XFLASH_MON_SEL [1:0]	Exposure prohibited control signal selection at short interval shutter 0h: 1 st exposure prohibited control signal 1h: 2 nd exposure prohibited control signal 2h: 1 st and 2 nd exposure prohibited control signal Others: setting prohibited	0h	00h	S	
		1						
		2	XFLASH_MON_INV [0]	Exposure prohibited control signal polarity selection 0: High active 1: Low active	0		S	
		3		Fixed to 0	0		—	
		4	EXP_MON_SEL [1:0]	Exposure period control signal selection at short interval shutter 0h: 1 st exposure period control signal 1h: 2 nd exposure period control signal 2h: 1 st and 2 nd exposure period control signal Others: setting prohibited	0h		S	
		5						
		6	EXP_MON_INV [0]	Exposure period control signal polarity selection 0: High active 1: Low active	0		S	
		7		Fixed to 0	0		—	
A8h	33A8h	0	XFLASH_MON1_1U [13:0]	LSB	0000h	00h	S	
		1						
		2						
		3						
		4						
		5						
		6		Timing setting of rise edge of the exposure prohibited control signal between the 1 st and 2 nd exposures ※ set the value of multiple of 4				
		7						
A9h	33A9h	0			0000h	00h	S	
		1						
		2						
		3						
		4						
		5		MSB				
		6		Fixed to 0			0	—
		7		Fixed to 0			0	—
AEh	33AEh	0	XFLASH_MON1_2D [13:0]	LSB	0000h	00h	S	
		1						
		2						
		3						
		4						
		5						
		6		Timing setting of fall edge of the exposure prohibited control signal between the 1 st and 2 nd exposures ※ set the value of multiple of 4				
		7						
AFh	33AFh	0			0000h	00h	S	
		1						
		2						
		3						
		4						
		5		MSB				
		6		Fixed to 0			0	—
		7		Fixed to 0			0	—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
B0h	33B0h	0	XFLASH_MON2_1U [13:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6					
B1h	33B1h	7		Timing setting of rise edge of the exposure prohibited control signal between the 2 nd exposure and next 1 st exposure. ※ set the value of multiple of 4			
		0			0	00h	—
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0			
B6h	33B6h	7		Fixed to 0			
		0	XFLASH_MON2_2D [13:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6		Timing setting of fall edge of the exposure prohibited control signal between the 2 nd exposure and next 1 st exposure. ※ set the value of multiple of 4			
B7h	33B7h	7					
		0			0	00h	—
		1					
		2					
		3					
		4					
		5		MSB			
		6		Fixed to 0			
B8h	33B8h	7		Fixed to 0			
		0	EXP_MON1_1U [13:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6		Timing setting of rise edge of the exposure period control signal at 1 st exposure. ※ set the value of multiple of 4			
B9h	33B9h	7					
		0			0	00h	—
		1					
		2					
		3					
		4					
		5		MSB			
		6					
		7					

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
BEh	33BEh	0	EXP_MON1_2D [13:0]	LSB	0000h	00h	S
		1		Timing setting of fall edge of the exposure period control signal at 1 st exposure. ※ set the value of multiple of 4			
		2					
		3					
		4					
		5					
		6					
7							
BFh	33BFh	0		MSB		00h	
		1					
		2					
		3					
		4					
		5					
		6					
7							
C0h	33C0h	0	EXP_MON2_1U [13:0]	LSB	0000h	00h	S
		1		Timing setting of rise edge of the exposure period control signal at 2 nd exposure. ※ set the value of multiple of 4			
		2					
		3					
		4					
		5					
		6					
7							
C1h	33C1h	0		MSB		00h	
		1					
		2					
		3					
		4					
		5					
		6					
7							
C6h	33C6h	0	EXP_MON2_2D [13:0]	LSB	0000h	00h	S
		1		Timing setting of fall edge of the exposure period control signal at 2 nd exposure. ※ set the value of multiple of 4			
		2					
		3					
		4					
		5					
		6					
7							
C7h	33C7h	0		MSB		00h	
		1					
		2					
		3					
		4					
		5					
		6					
7							

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3400h	0	TRIGMODE [2:0]	Global shutter mode setting 0h: Normal mode 4h: Short Interval Shutter mode Others: setting prohibited	0h	00h	S
		1					
		2					
		3	TRIGTIMING [1:0]	Fixed to 0	0h		S
		4					
		5	Fixed to 0	0	—		
		6	Fixed to 0	0	—		
		7	Fixed to 0	0	—		
36h	3436h	0	TOUT0SEL [2:0]	TOUT0 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal Others: setting prohibited	0h	00h	S
		1					
		2					
		3	TOUT1EXPSEL [2:0]	TOUT0 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal Others: setting prohibited	0h		S
		4					
		5					
		6	TOUT2EXPSEL [2:0]	TOUT0 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal Others: setting prohibited	0h		S
		7					
37h	3437h	0		Others: setting prohibited		00h	
		1	Fixed to 0	0	—		
		2	Fixed to 0	0	—		
		3	Fixed to 0	0	—		
		4	Fixed to 0	0	—		
		5	Fixed to 0	0	—		
		6	Fixed to 0	0	—		
		7	Fixed to 0	0	—		

Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing							
4-wire	I ² C				By register	By address								
02h	3502h	0	GAIN_RTS [7:0]	Setting of Gain Reflection Timing. Fixed o 8h at the short interval shutter mode	00h	00h	S							
		1												
		2												
		3												
		4												
		5												
		6												
		7												
		0						GAIN [8:0]	FID: 0 gain setting	000h	00h	V		
		1												
		2												
		3												
		4												
		5												
6														
		7												
		0						MSB						
		1						Fixed to 0						
		2						Fixed to 0						
		3						Fixed to 0						
		4						Fixed to 0						
		5						Fixed to 0						
6	Fixed to 0													
		7	Fixed to 0	0										
		0	GAIN1 [8:0]	FID: 1 gain setting				000h	00h	V				
		1									LSB			
		2									FID: 1 gain setting			
		3												
		4												
		5												
6														
7														
		0												
		1						MSB						
		2						Fixed to 0						
		3						Fixed to 0						
		4						Fixed to 0						
		5						Fixed to 0						
		6						Fixed to 0						
7	Fixed to 0													
		0												
		1						Fixed to 0						
		2						Fixed to 0						
		3						Fixed to 0						
		4						Fixed to 0						
		5						Fixed to 0						
		6						Fixed to 0						
7	Fixed to 0													

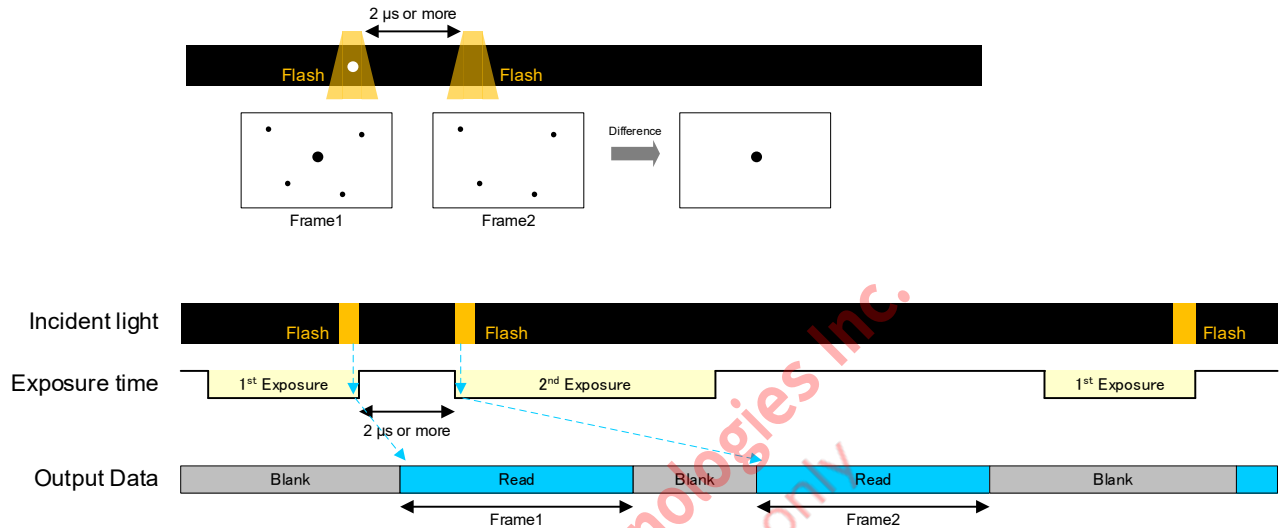
5. Short Interval Shutter mode

5.1. Overview

This function performs twice exposure at shutter intervals of $2\ \mu\text{s}$ or more.

It is assumed to be used to expose two consecutive frames to exposure lights respectively in a short interval, for example, for Particle Image Velocimetry (PIV) which is used for detection of particles or measurement of particle velocity. The 1st exposure period, the 2nd exposure period, and the interval between 1st and 2nd exposures (exposure prohibition period) can be adjusted by the register (unit: 13.47 ns).

For exposure period and exposure prohibited period, control signals can be monitored using TOUT0 to TOUT2.



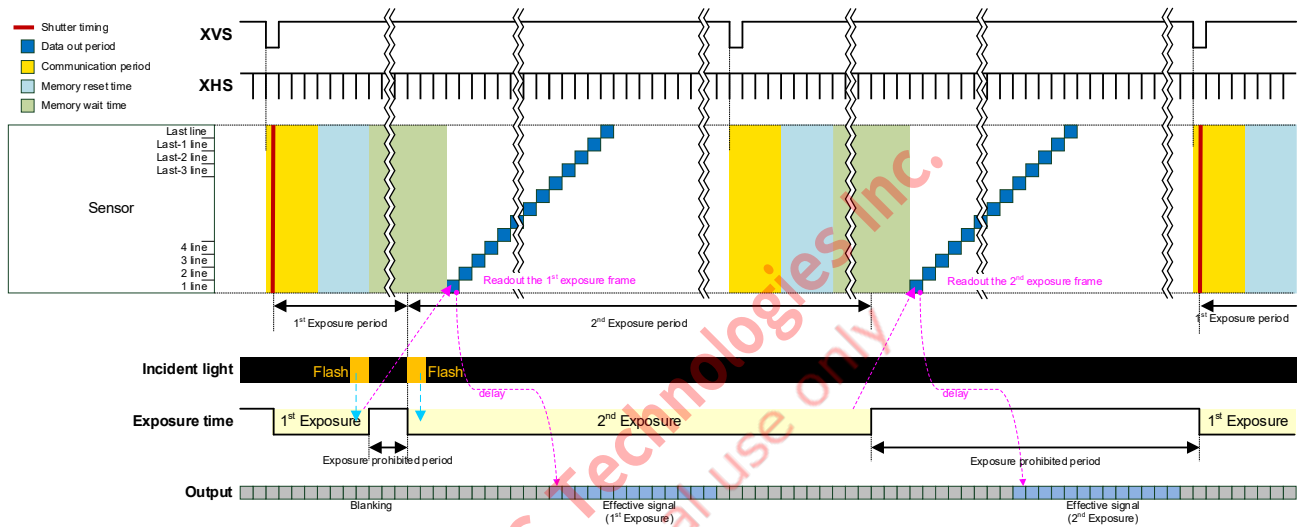
Output Image of Short Interval Shutter

6. Setting of Short Interval Shutter

By setting 4h to the global shutter mode register (TRIGMODE [2: 0]), this sensor will be in the short interval shutter mode. This function operates as one set (1 Frameset) with two XVS inputs.

List of Global Shutter mode setting

Register	Register details			Initial value	Description	Note
	Chip ID	Address (I ² C)	bit			
TRIGMODE [2:0]	06h	00h (3400h)	[2:0]	0h	0h: Normal mode 4h: Short Interval Shutter mode	Fixed to 0 at the short interval shutter mode
TRIGTIMING [1:0]			[4:3]	0h	0h	



Drive timing of Short Interval Shutter function

6.1. Exposure Period

The 1st exposure period is the Communication period, Memory reset time, and Memory wait time after the first XVS input. The 2nd exposure period is from Memory wait time after the 1st XVS input to Memory wait time after the 2nd XVS input. The exposure period can be changed (unit: H) by setting the following registers.

List of Exposure period registers

Register	Register details			Initial value			Description	Note
	Chip ID	Address (): I ² C	bit					
VNDMY [7:0]	02h	E0h (30E0h)	[7:0]	04h			Communi- cation period (Unit: H)	1 st exposure period *1
GMRWT [7:0]		E2h (30E2h)	[7:0]	0Ch*2	10h*3	06h*4	Memory reset time period (Unit: H)	
GMTWT [7:0]		E3h (30E3h)	[7:0]	48h*2	70h*3	28h*4	Memory wait time period (Unit: H)	
VMAX [23:0]		D4h (30D4h)	[7:0]	0012ECh*2	000CD4h*3	0008AAh*4	2 nd exposure period (Unit: H)	
		D5h (30D5h)	[7:0]					
		D6h (30D6h)	[7:0]					

^{*1} To extend the first exposure period, change VNDMY [7:0].

GMRWT [7:0] and GMTWT [7:0] have to set at the setting value in each drive mode.

^{*2} IMX530 series and IMX540 series

^{*3} IMX535 series, IMX545 series and IMX565 series

^{*4} IMX548 series and IMX568 series

6.2. Exposure Interval and Exposure period

The exposure period and exposure interval of the short interval shutter mode are controlled by the exposure period control signal and the exposure prohibited control signal.

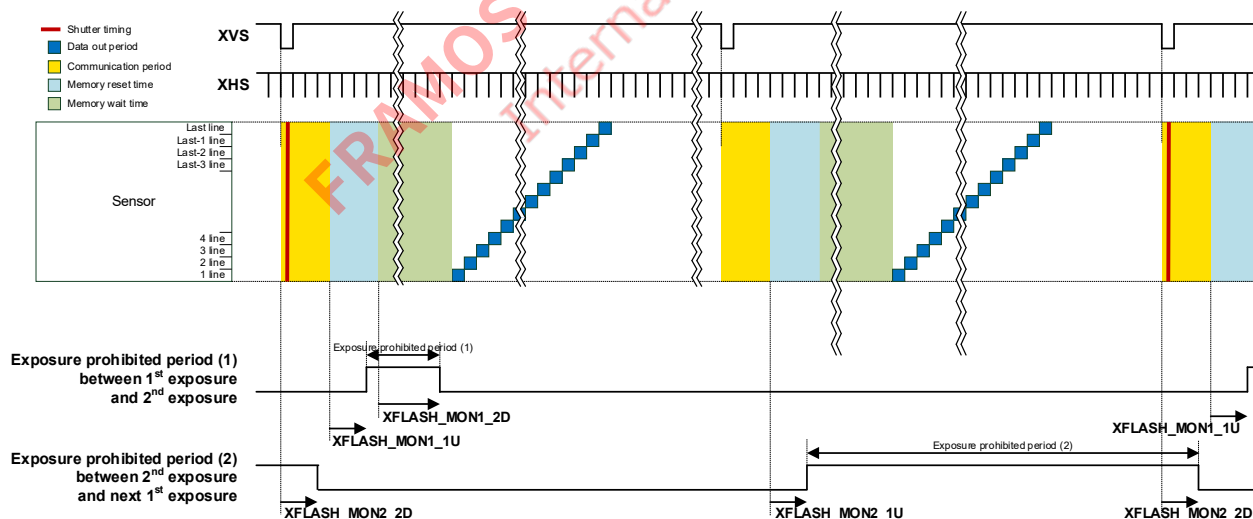
6.2.1. Exposure prohibited control signal

Control the exposure prohibited period at the short interval shutter mode.

The exposure interval between the 1st exposure and the 2nd exposure (exposure prohibited period (1)) and the exposure interval between the 2nd exposure and the 1st exposure of the next frameset (exposure prohibited period (2)) can be adjusted (unit: 13.47 ns) by the following registers.

List of Exposure prohibited control registers

Register	Register details			Initial value	Description	Note
	Chip ID	Address (): I ² C	bit			
XFLASH_MON1_1U [13:0]	05h	A8h (33A8h)	[7:0]	0000h	Rise edge of the exposure prohibited period (1)	Set within the 1 st Memory reset time period.
		A9h (33A9h)	[5:0]			
XFLASH_MON1_2D [13:0]		AEh (33AEh)	[7:0]	0000h	Fall edge of the exposure prohibited period (1)	Set within the 1 st Memory wait time period.
		AFh (33AFh)	[5:0]			
XFLASH_MON2_1U [13:0]		B0h (33B0h)	[7:0]	0000h	Rise edge of the exposure prohibited period (2)	Set within the 2 nd Memory reset time period.
		B1h (33B1h)	[5:0]			
XFLASH_MON2_2D [13:0]		B6h (33B6h)	[7:0]	0000h	Fall edge of the exposure prohibited period (2)	Set within the Communication period of next frameset
		B7h (33B7h)	[5:0]			



Exposure prohibited control

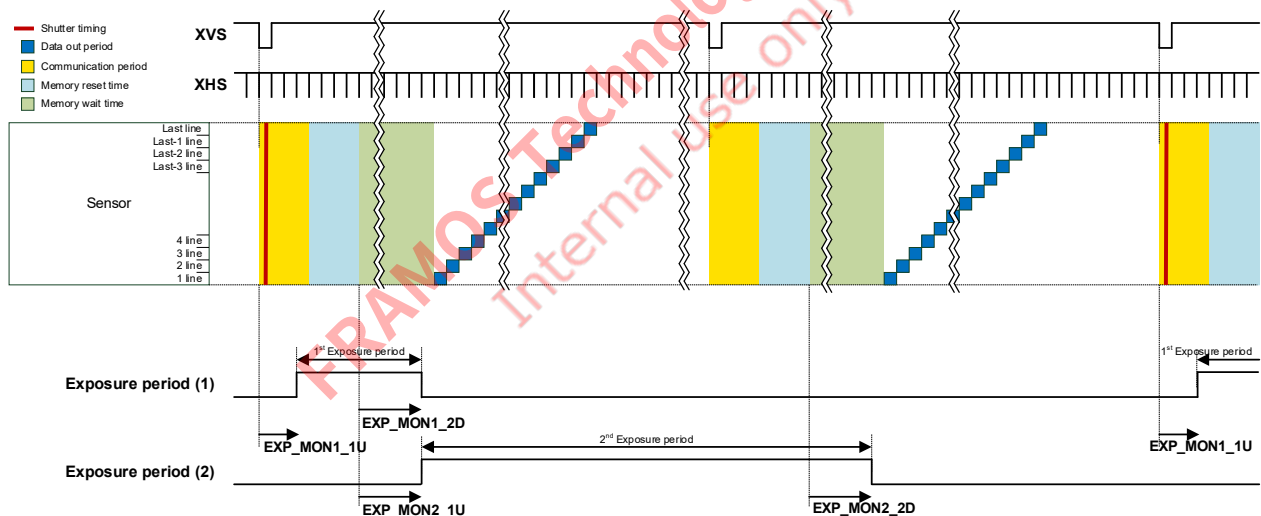
6.2.2. Exposure period control signal

Controls the exposure time for the short interval shutter mode.

Adjust the first exposure time (exposure time (1)) and the second exposure time (exposure time (2)) by the following registers. (Unit: 13.47 ns)

List of Exposure period control registers

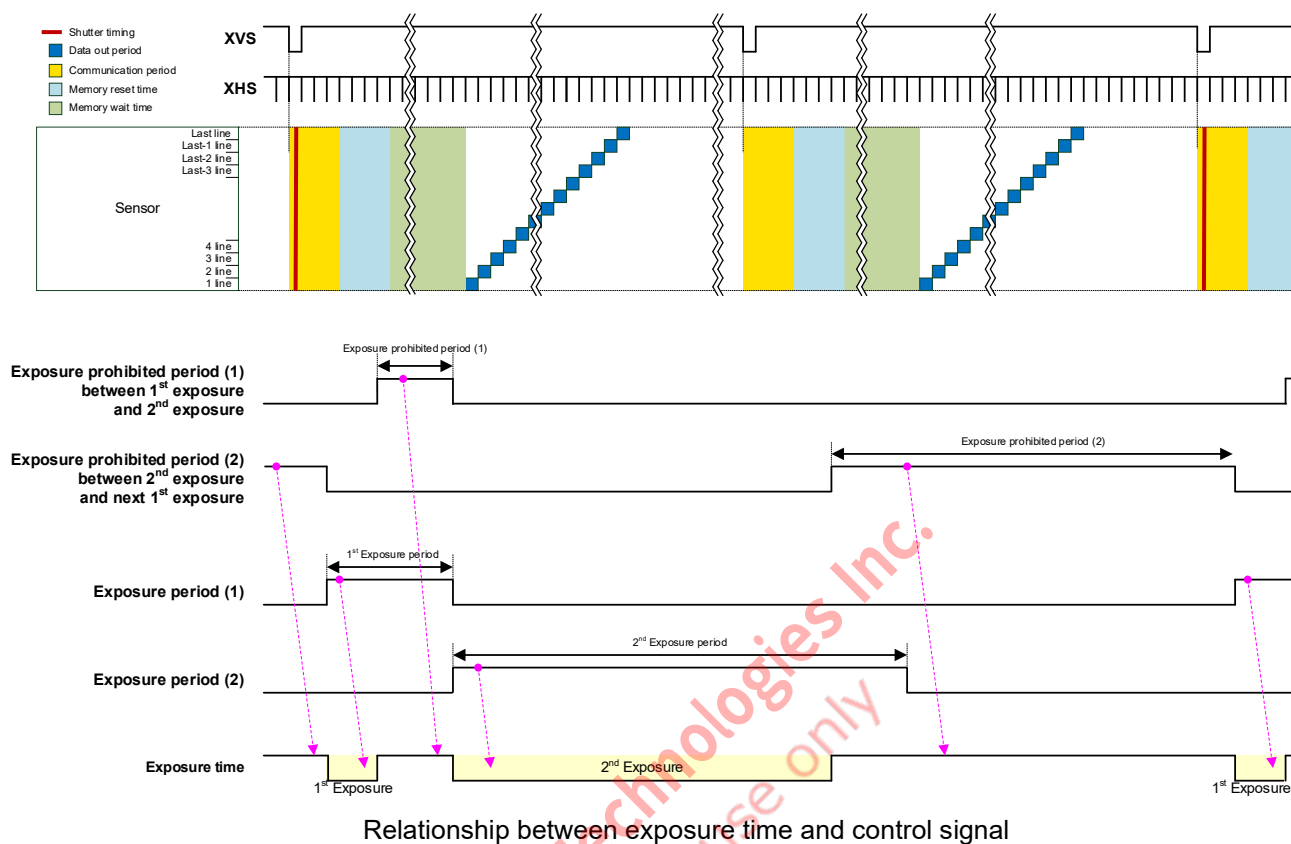
Register	Register details			Initial value	Description	Note
	Chip ID	Address (): I ² C	bit			
EXP_MON1_1U [13:0]	05h	B8h (33B8h)	[7:0]	0000h	Rise edge of exposure period control signal within 1 st exposure period	Set within the 1 st communication period.
		B9h (33B9h)	[5:0]			
EXP_MON1_2D [13:0]		BEh (33BEh)	[7:0]	0000h	Fall edge of exposure period control signal within 1 st exposure period	Set within the 1 st Memory wait time period.
		BFh (33BFh)	[5:0]			
EXP_MON2_1U [13:0]		C0h (33C0h)	[7:0]	0000h	Rise edge of exposure period control signal within 2 nd exposure period	Set within the 1 st Memory wait time period.
		C1h (33C1h)	[5:0]			
EXP_MON2_2D [13:0]		C6h (33C6h)	[7:0]	0000h	Fall edge of exposure period control signal within 2 nd exposure period	Set within the 2 nd Memory wait time period.
		C7h (33C7h)	[5:0]			



Exposure period control

6.2.3. Relationship between exposure time and control signals

The relationship between exposure time and control signals (exposure period control signals, exposure prohibited control signals) is as follows.



6.2.4. Pulse Output Function

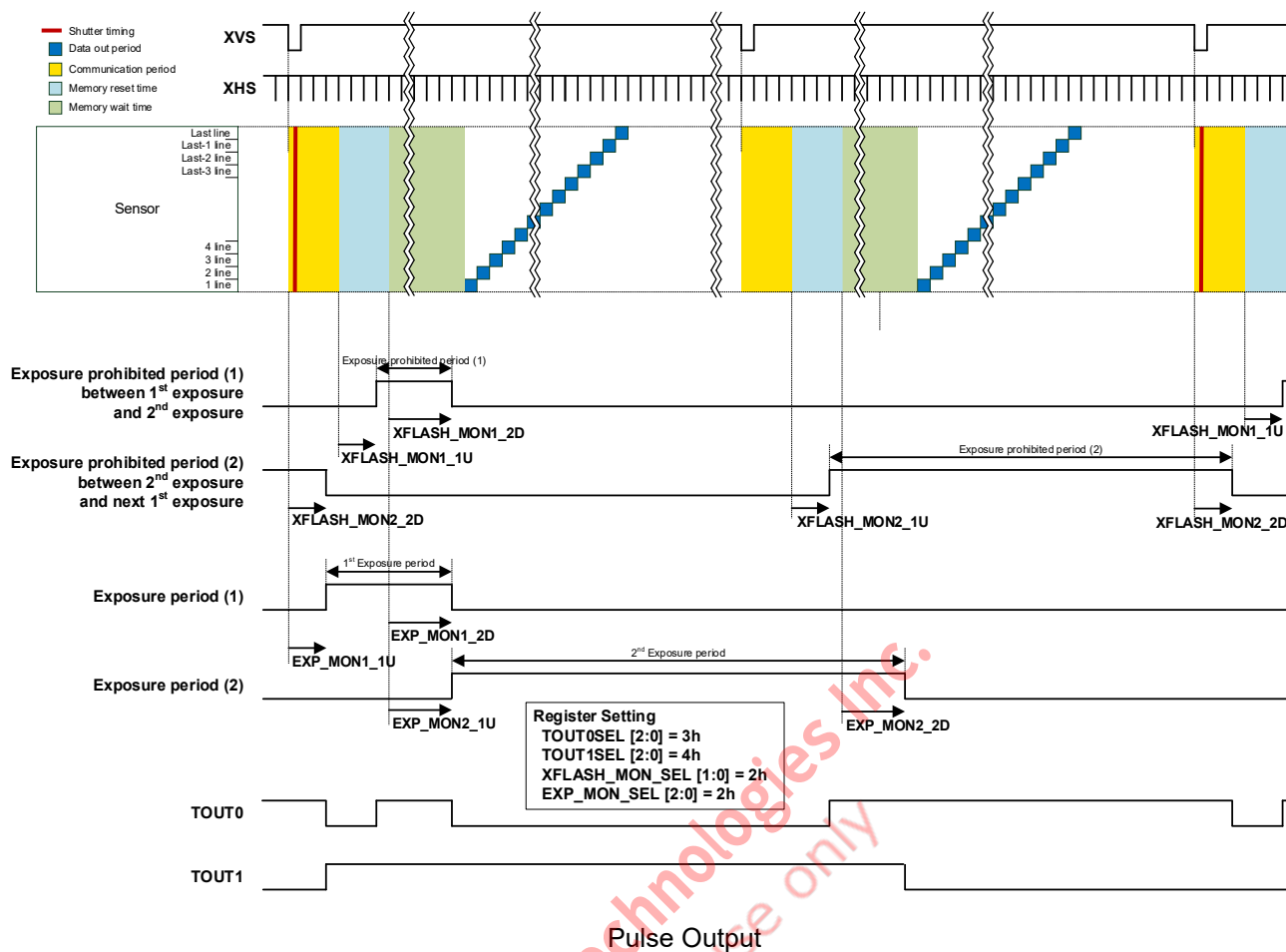
This sensor can output the exposure period control signal and the exposure prohibited control signal to TOUT0, TOUT1 and TOUT2 during the short interval shutter mode.

The control signals that can be pulse output can be selected from the following.

- 1st exposure period control signal
- 2nd exposure period control signal
- 1st and 2nd exposure period control signal
- 1st exposure prohibited control signal (exposure interval between 1st exposure and 2nd exposure)
- 2nd exposure prohibited control signal (exposure interval between 2nd exposure and the 1st exposure of the next frameset)
- 1st and 2nd exposure prohibited control signals

List of Pulse output function registers

Register	Register details			Initial value	Description
	Chip ID	Address (I ² C)	bit		
XFLASH_MON_SEL [1:0]	05h	A4h (33A4h)	[1:0]	0h	Selection of exposure prohibited period signals 0h: 1 st exposure prohibited control signal 1h: 2 nd exposure prohibited control signal 2h: 1 st and 2 nd exposure prohibited control signal
XFLASH_MON_INV [0]			[2]	0	Exposure prohibited control signal polarity selection 0: High active 1: Low active
EXP_MON_SEL [1:0]			[5:4]	0h	Selection of exposure time period signals 0h: 1 st exposure period control signal 1h: 2 nd exposure period control signal 2h: 1 st and 2 nd exposure period control signal
EXP_MON_INV [0]			[6]	0	Exposure period control signal polarity selection 0: High active 1: Low active
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal
TOUT1EXPSEL [2:0]			[5:3]	0h	TOUT1 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal
TOUT2EXPSEL [2:0]		37h (3437h)	[0]	0h	TOUT2 pin setting 0h: Low fixed 3h: Exposure prohibited control signal 4h: Exposure period control signal



7. ROI Function

7.1. In case of ROI

The short interval shutter mode can be used with ROI function.

The ROI settings are basically the same to the data sheet. Refer to the data sheet for details.

Cropping position can set maximum 16 areas that specified by horizontal 4 points and vertical 4 points, regarding effective pixel start position as origin (0, 0) in all pixel scan mode.

Also, for the short interval shutter modes, the ROI areas of the first exposure frame and the second exposure frame are individually it can be set.

When the number of vertical lines is different between the first exposure frame and the second exposure frame, the VMAX (in master mode) and the XVS input timing (in slave mode) have to be set according to the frame with the largest number of lines.

Refer to the product specifications for the number of lines in the frame.

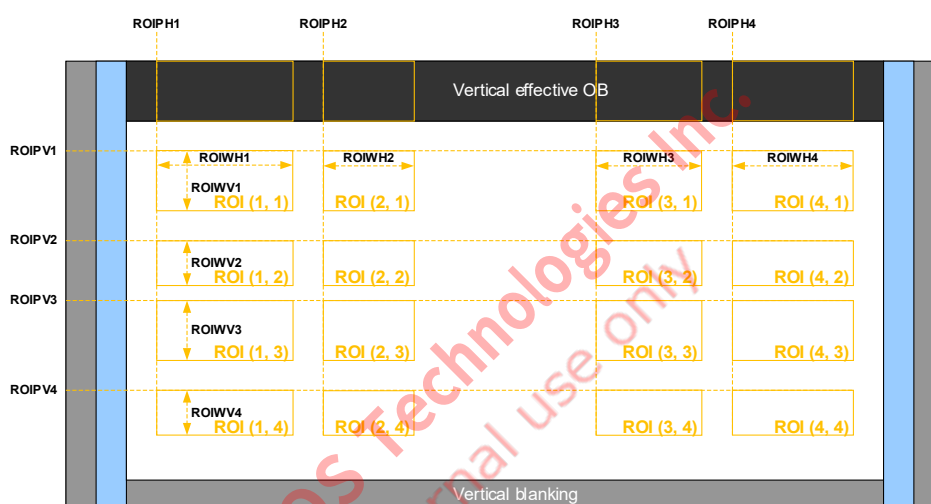
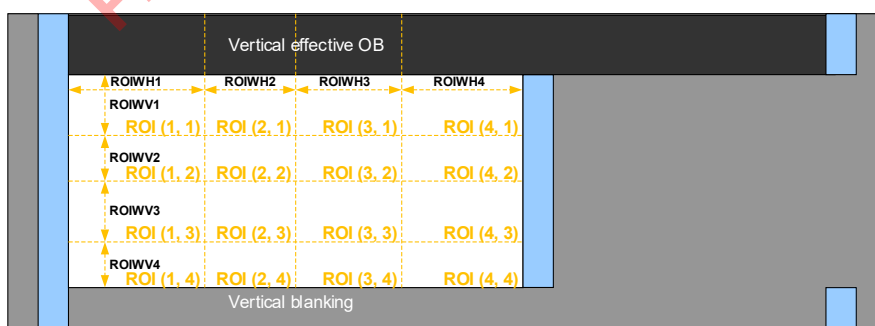


Image Drawing of Designated Areas in ROI Mode (SLVS output)



Details of Image Drawing (SLVS output)

- ◆ The short interval shutter mode is able to set the different ROI area for each frame.

The register settings should satisfy following conditions:

- * Do not designate area like be overlap.

$$\text{ROIPH1} + \text{ROIWH1} \leq \text{ROIPH2}$$

$$\text{ROIPH2} + \text{ROIWH2} \leq \text{ROIPH3}$$

$$\text{ROIPH3} + \text{ROIWH3} \leq \text{ROIPH4}$$

$$\text{ROIPH4} + \text{ROIWH4} \leq \text{The horizontal width at All-pixel scan.}$$

$$\text{ROIPV1} + \text{ROIWV1} \leq \text{ROIPV2}$$

$$\text{ROIPV2} + \text{ROIWV2} \leq \text{ROIPV3}$$

$$\text{ROIPV3} + \text{ROIWV3} \leq \text{ROIPV4}$$

$$\text{ROIPV4} + \text{ROIWV4} \leq \text{The vertical width at All-pixel scan.}$$

- * Set the horizontal and vertical setting in multiple of 16.

(IMX530 series, IMX535 series)

- * Set the horizontal and vertical setting in multiple of 8.

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

- * The range of horizontal output width is as below.

10 / 12 bit mode

$$\text{ROIWH1} + \text{ROIWH2} + \text{ROIWH3} + \text{ROIWH4} \geq 608d$$

8 bit mode

$$\text{ROIWH1} + \text{ROIWH2} + \text{ROIWH3} + \text{ROIWH4} \geq 1200d$$

- * The range of vertical output width is as below.

8 / 10 / 12 bit mode

$$\text{ROIWV1} + \text{ROIWV2} + \text{ROIWV3} + \text{ROIWV4} \geq 16d$$

$$\text{ROIWV1} + \text{ROIWV2} + \text{ROIWV3} + \text{ROIWV4} \geq 8d$$

(IMX530 series, IMX535 series)

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

Refer to "Reflection Timing" for reflection timing of FID.

7.2. In case of Overlap ROI

The setting of ROI is basically similar to the datasheet. Refer to the datasheet for details.

Cropping position can set maximum 4 areas that specified effective pixel start position as origin (0, 0) in all pixel scan mode.

When the number of lines in the vertical direction is different between the 1st exposure frame and the second exposure frame, the VMAX (in master mode) and XVS input timing (in slave mode) have to be set according to the largest frame.

Refer to the product specifications for the number of lines in the frame.

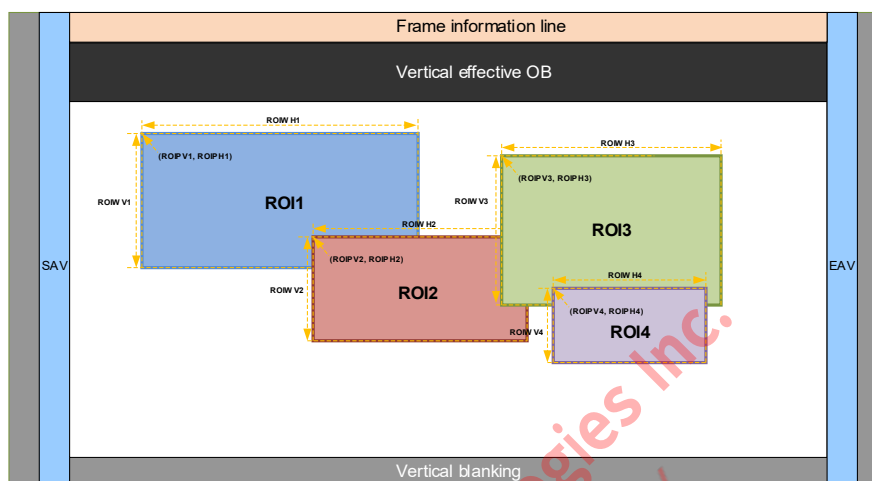
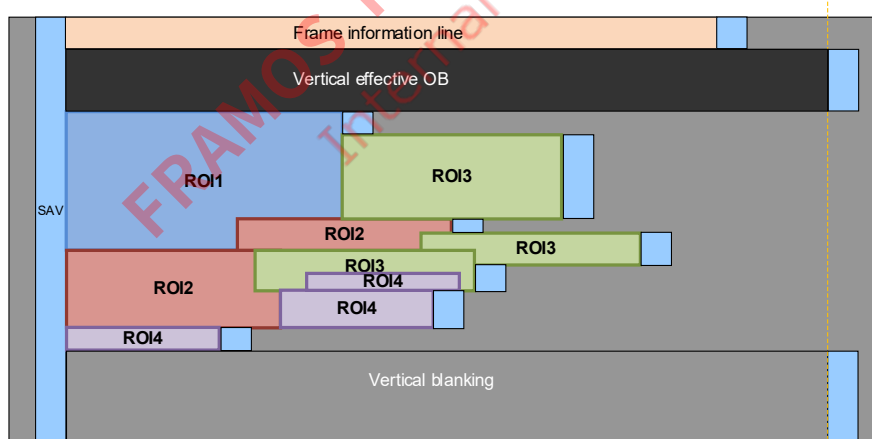


Image Drawing of Designated Areas in Overlap ROI Mode (SLVS output)



Details of Image Drawing (SLVS output)

- ◆ The short interval shutter mode is able to set the different ROI area for each frame.

The register settings should satisfy following conditions:

$ROI_{PH1} + ROI_{WH1} \leq$ The horizontal width at All-pixel scan
 $ROI_{PH2} + ROI_{WH2} \leq$ The horizontal width at All-pixel scan
 $ROI_{PH3} + ROI_{WH3} \leq$ The horizontal width at All-pixel scan
 $ROI_{PH4} + ROI_{WH4} \leq$ The horizontal width at All-pixel scan.

$ROI_{PV1} + ROI_{VW1} \leq$ The vertical width at All-pixel scan
 $ROI_{PV2} + ROI_{VW2} \leq$ The vertical width at All-pixel scan
 $ROI_{PV3} + ROI_{VW3} \leq$ The vertical width at All-pixel scan
 $ROI_{PV4} + ROI_{VW4} \leq$ The vertical width at All-pixel scan.

* Set the horizontal and vertical setting in multiple of 16.

(IMX530 series, IMX535 series)

* Set the horizontal and vertical setting in multiple of 8.

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

* The range of horizontal output width is as below.

10 / 12 bit mode

$ROI_{WH1} + ROI_{WH2} + ROI_{WH3} + ROI_{WH4} \geq 608d$

8 bit mode

$ROI_{WH1} + ROI_{WH2} + ROI_{WH3} + ROI_{WH4} \geq 1200d$

* The range of vertical output width is as below.

8 / 10 / 12 bit mode

$ROI_{VW1} \geq 16d, ROI_{VW2} \geq 16d, ROI_{VW3} \geq 16d, ROI_{VW4} \geq 16d$

$ROI_{VW1} \geq 8d, ROI_{VW2} \geq 8d, ROI_{VW3} \geq 8d, ROI_{VW4} \geq 8d$

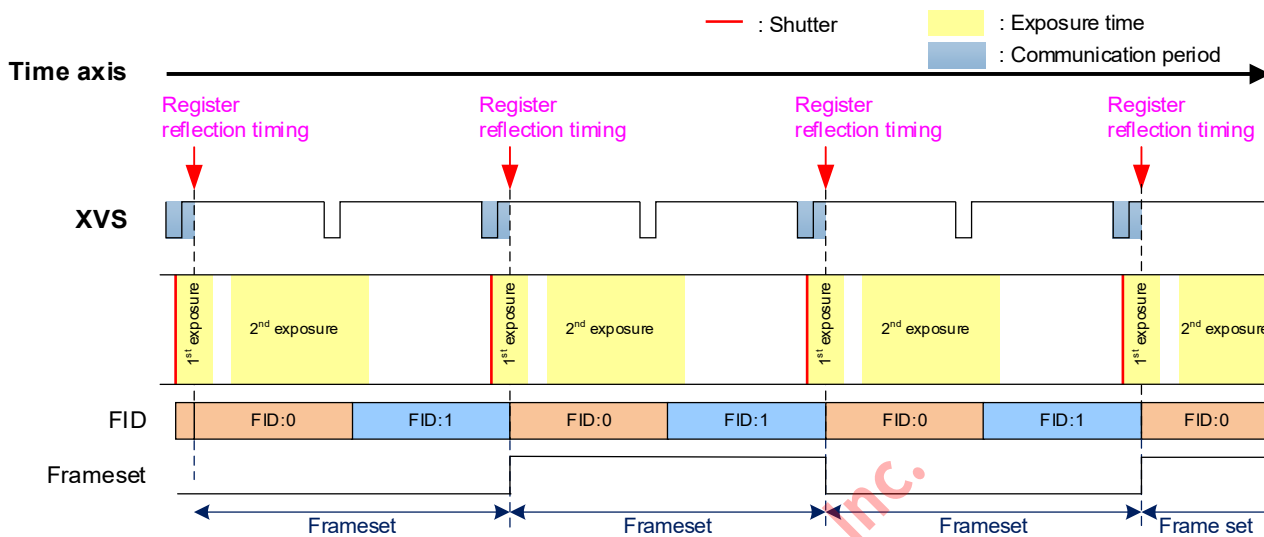
(IMX530 series, IMX535 series)

(IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series)

Refer to "Reflection Timing" for reflection timing of FID.

8. Reflection Timing

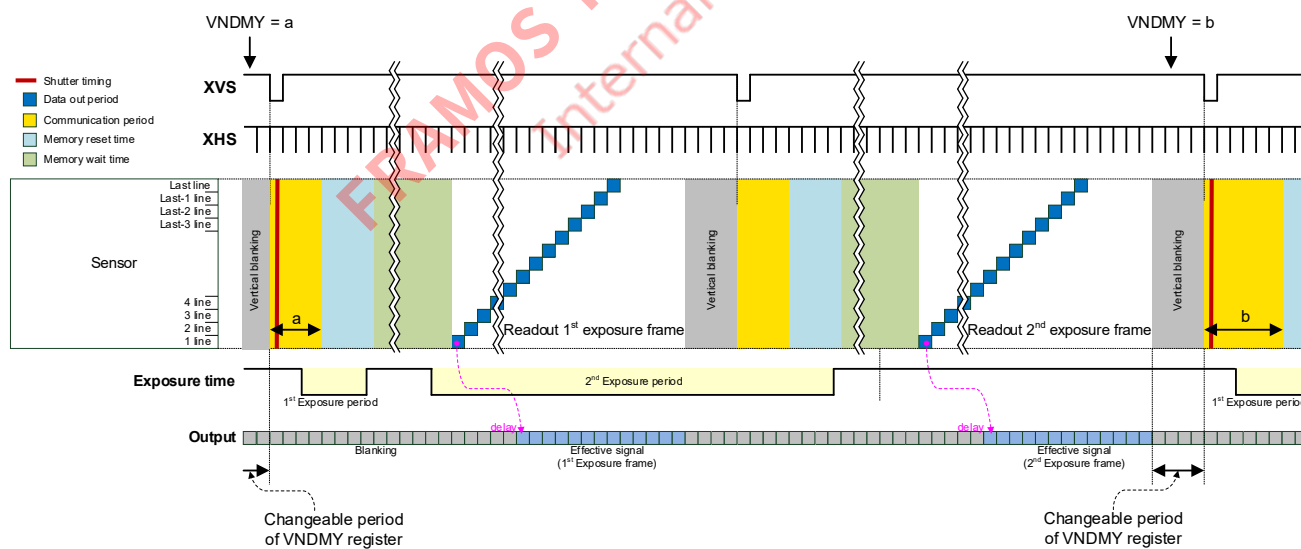
In the short interval shutter mode, the registers are reflected at the timing as shown in the figure below (immediately after the communication period by the first XVS). It will be reflected in 1 frame set unit (2XVS).



Register Reflection Timing at Short Interval Shutter

8.1. Timing of the 1st exposure period change

When adjusting the 1st exposure period (VNDMY [7:0]) of Frameset, change the setting during the 2nd Vertical blanking period of the previous Frameset.



Timing of 1st Exposure period register setting

9. Each frame Gain mode

Set an individual gain for each exposure frame. The setting STEP is 0.1 dB.

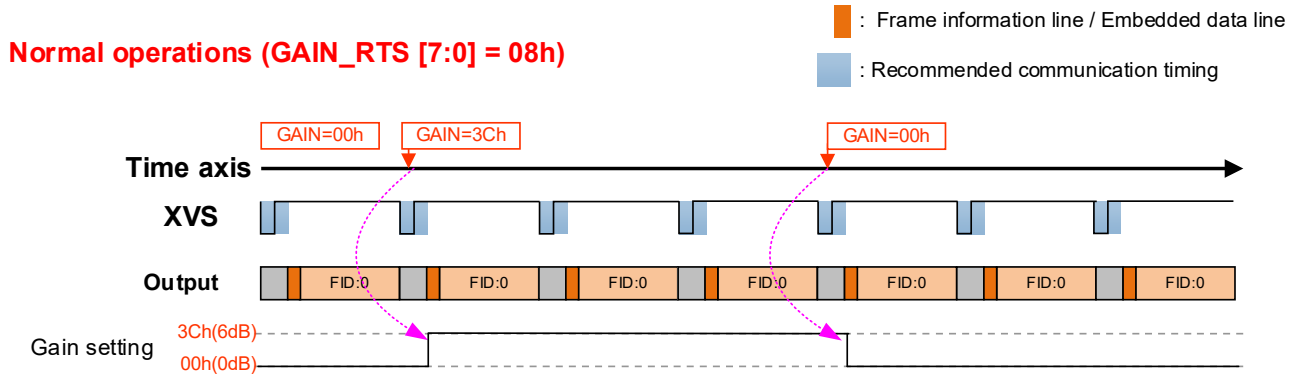
List of each frame gain registers

Register details			Register	Setting value		Description
Chip-ID	Address (): I ² C	bit		Normal mode	Short Interval Shutter mode	
07h	14h (3514h)	[7:0]	GAIN [8:0]	Gain setting 0 to 48 dB	FID: 0 Gain setting 0 to 48 dB	Gain setting
	15h (3515h)	[0]				
	16h (3516h)	[7:0]	GAIN1 [8:0]	Fixed to "000h"	FID: 1 Gain setting 0 to 48 dB	Gain setting
	17h (3517h)	[0]				

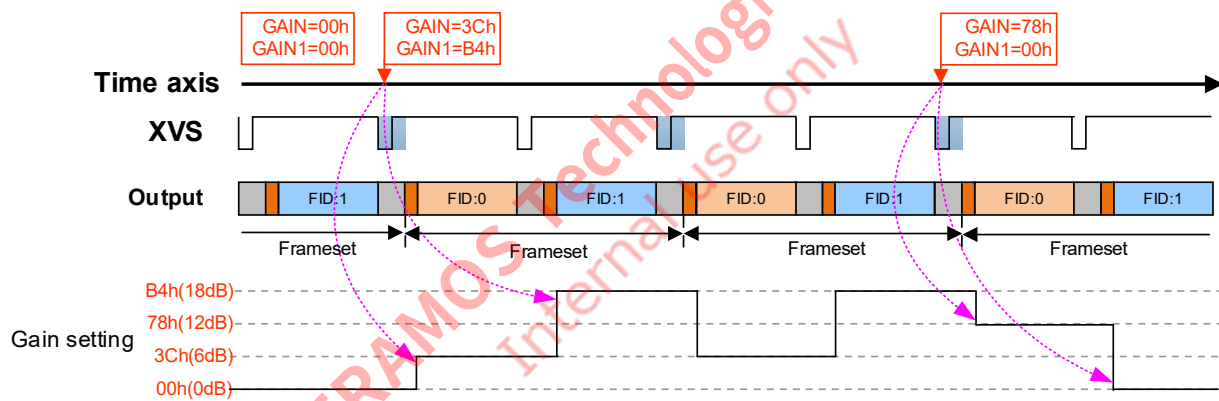
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9.1. Gain reflection timing

In the short interval shutter mode, the reflection timing of gain setting is reflected collectively to the unit of frameset. The gain reflection timing is shown figure below. (In the short interval shutter mode, set 08h to GAIN_RTS [7:0] register (Chip-ID = 07h, Address = 02h).)



Short Interval Shutter (GAIN_RTS [7:0] = 08h)



Gain Reflection Timing

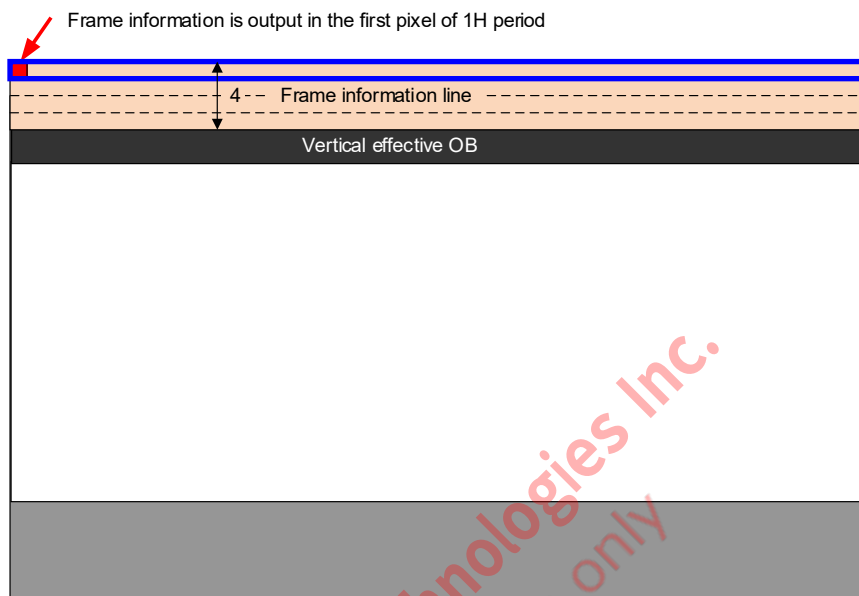
10. Frame identification method

The identification signal of each frame is Frame ID, and the unit of the short interval shutter is Frameset.

10.1. Frame information line

This product uses the Frame information line of SLVS to output Frame ID and Frame set.

Although 4 lines of the Frame information line are output in this product, it is added to the output signal of the 1st pixel of the blue frame in the figure below.



Frame information line output position (In case of SLVS)

10.2. Output format (Frame information line)

The output format of Frame information line is shown below.

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

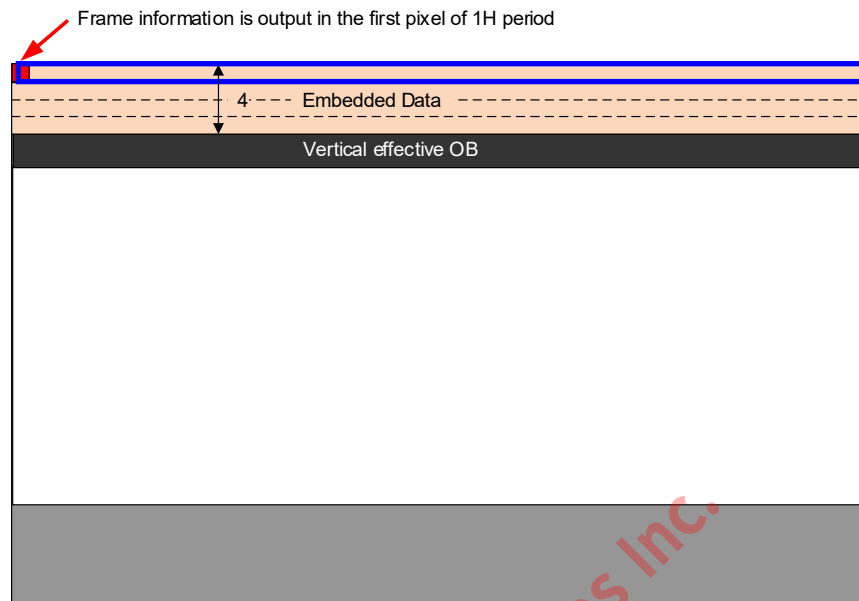
Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Frame information line)

10.3. Embedded data line

This product uses the Embedded data line of SLVS – EC to output Frame ID and Frame set.

Although 4 lines of the Embedded data line are output in this product, it is added to the output signal of the 1st pixel of the blue frame in the figure below.



Embedded data line output position (In case of SLVS – EC)

10.4. Output format (Embedded data line)

The output format of Frame information line is shown below.

(SLVS-EC)

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Embedded data line (SLVS-EC))

(CSI-2)

Output	8bit
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	Frame ID [1]
[8]	Frame ID [0]
[7]	0
[6]	0
[5]	0
[4]	Frameset
[3]	Frame ID [1]
[2]	Frame ID [0]
[1]	0
[0]	1

Output	12bit
[11]	Frame ID [1]
[10]	Frame ID [0]
[9]	0
[8]	0
[7]	0
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	1
[1]	0
[0]	1

Output format (Embedded data line (CSI-2))

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11. Mode Transitions

The transition related to the short interval shutter mode is shown below. If perform the transition which is not described in table below, set via sensor standby.

11.1. Mode Transitions

Mode Transition			Note
Change of operation mode			
Normal mode	→	Short Interval Shutter mode	Set via the sensor standby.
Short Interval Shutter mode	→	Normal mode	
Change of driving mode			
All-pixel	→	ROI	There can transition without invalid frame. Changing of the registers FREQ and HMAX is prohibited. When changing the frame rate (VMAX change), one invalid frame is generated.
ROI	→	All-Pixel	
ROI	→	ROI	

- The exposure period for 1st frame during the short interval shutter mode can be changed without invalid frame.

12. Limitations

- It does not support driving in Trigger mode.
- It does not support driving with Dual ADC mode.
- It does not support driving with Multi Frame set output mode.
- It does not support driving with Self Trigger mode.
- Driving with the combination of "1 / 2 Subsampling mode" and "ROI mode" is not supported.
- Driving with the combination of "1 / 2 Subsampling mode" and "Overlap ROI mode" is not supported.

Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 4 / 26	–	First Edition
Rev.0.2	2019 / 6 / 28	–	Change the file name and the function name from Twice Exposure to Short Interval Shutter
		3, 15, 18, 19, 20, 21, 22, 23, 24, 26, 27, 29, 30, 31, 32, 33, 35	Change the function name from Twice Exposure to Short Interval Shutter
		35	Added a note on mode transition
Rev.0.3	2019 / 8 / 30	1	Change “1. Description” Add the sensor types to 2 × 2 FD binning in “2. Features”
		1, 3, 15, 16, 17, 29, 22, 23, 25, 36	Change to the exposure period control signal from exposure time control signal
		3	Change the reflection timing of VNDMY register to “I” Added the constraint for VNDMY setting
		20	Change “4.1 Overview”
		20, 21, 22, 23, 26, 31, 33	update the figure
		23	Added setting range to note of exposure period control related registers
		24	Added “5.2.3. Relationship between exposure time and control signals”
Rev.1.0	2020 / 1 / 7	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added: Product types (IMX531, IMX532, IMX540, IMX541, IMX542) • Added: Chapter “Product type” Update: TBD in the chapter “Description” • “2 μs or more (TBD)” → “2 μs or more” Correction: 2x2 FD binning mode • Correction: “Only IMX530-AAMJ-C” → “The product with a color filter is excluded” Added: SLVS output (2 ch) in the support interface
Rev.2.0	2021 / 3 / 4	1, 2	Added: IMX535 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series
		2, 38	Added: CSI-2 serial data output
		4	Correction: Reflection timing VNDMY [7:0] “I” → “S”
		4, 5	Correction: “The horizontal setting” → “The vertical setting”
		5 to 15 30, 32	Added: Registers value of IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series
		23	Correction: List of Exposure period register

Version	Date	Page	Remarks
Rev.2.0	2021 / 3 / 4	30	Correction: ROI specified setting conditions "≤" → "≤"
		32	Correction: Expression of "The range of vertical output width"
		40	Added: Two items about combination drive to "12. Limitations"
Rev.3.0	2022 / 5 / 11	20, 34, 35	Delete: WD_GAIN_MODE

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